Network Science

Lecture 05: Social Context and Link Formation

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This lecture extends the network model with context, attributes, and longitudinal processes.

Speaker notes

This lecture brings social context into the network model. So far the graph has been an abstract object with nodes and edges. Now we add:

- attributes on nodes (homophily)
- mechanisms of formation (selection vs. socialization)
- a second node type for foci (affiliation networks)
- temporal observations of edge formation

The pedagogical anchor is that links are never neutral data points — they reflect choices and contexts that the analyst must reconstruct.

Social Context and Link Formation

This lecture extends the network model with context, attributes, and longitudinal processes.

Focus

- homophily
- selection and socialization
- affiliation networks and closure beyond person-person ties
- online link formation and spatial segregation

Module 1

Homophily

Guiding Question: Do people become friends because they are similar, or do they become similar because they are friends?

Formal Intuition

Homophily means that edge formation is statistically biased toward similarity in node attributes.

Speaker notes

Introduce the structural idea before the example. The key point is that homophily is a bias in tie formation, not just a visual pattern.

Visual Intuition

Homophily as similarity-biased tie formation

Solid edges are frequent within-group ties; dashed edges are rarer cross-group ties.

Interpretation

Friendship ties are often biased toward similarity in attributes such as age, background, beliefs, or activity.

Why This Matters

Why This Matters

- it shapes who meets whom

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- it shapes who meets whom
- it changes which information spreads where

Why This Matters

- it shapes who meets whom
- it changes which information spreads where
- it complicates causal interpretation of network data

Quick Check 5.A: Detecting Homophily

In a school of 200 students, 60% are in track A and 40% in track B. Of all friendships, 78% are within-track.

1. What fraction of friendships would you expect to be within-track if friendships were random with respect to track?
2. Is the observed 78% evidence of homophily? Why or why not?

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Speaker notes

Give them 4 minutes. (1) Random probability of within-track friendship is $0.6^2 + 0.4^2 = 0.36 + 0.16 = 0.52$. (2) Yes — observed 78% is much higher than the random baseline 52%, suggesting homophily on track. The point is that "many same-group friendships" is not enough — you need a baseline.

Quick Check 5.A: Solution

Yes. Observed 78% is well above the 52% random baseline, indicating that friendships are biased toward within-track. The lesson: declaring homophily requires a counterfactual — "more than expected if links were random with respect to the attribute".

Quick Check 5.A: Solution

1. Random baseline:

Yes. Observed 78% is well above the 52% random baseline, indicating that friendships are biased toward within-track. The lesson: declaring homophily requires a counterfactual — "more than expected if links were random with respect to the attribute".

Quick Check 5.A: Solution

1. Random baseline:

$$P(\text{same track}) = 0.6^2 + 0.4^2 = 0.36 + 0.16 = 0.52$$

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Quick Check 5.A: Solution

1. Random baseline:

$$P(\text{same track}) = 0.6^2 + 0.4^2 = 0.36 + 0.16 = 0.52$$

2. Evidence of homophily: Yes. Observed 78% is well above the 52% random baseline, indicating that friendships are biased toward within-track. The lesson: declaring homophily requires a counterfactual — "more than expected if links were random with respect to the attribute".

Takeaway

> Homophily is not only a descriptive pattern. It is a mechanism that shapes network formation and measurement.

Module 2

Selection and Socialization

> **Guiding Question:** How can we distinguish selection from social influence?

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> **Guiding Question:** How can we distinguish selection from social influence?

$$\text{homophily} = \text{selection} + \text{socialization} + \text{context}$$

Core Distinction

- : people choose similar others
- : people become more similar after linking
- : both may reflect a third factor (shared context)

Speaker notes

Keep the causal distinction explicit: similarity at one point in time does not identify the mechanism that produced it.

Core Distinction

- **selection**: people choose similar others
- : people become more similar after linking
- : both may reflect a third factor (shared context)

Core Distinction

- **selection**: people choose similar others
- **socialization**: people become more similar after linking
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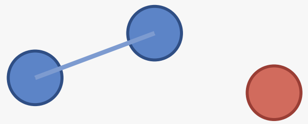
Core Distinction

- **selection**: people choose similar others
- **socialization**: people become more similar after linking
- **correlation**: both may reflect a third factor (shared context)

Example

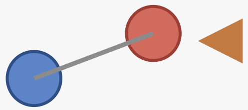
Why observed similarity can arise in different ways

Selection



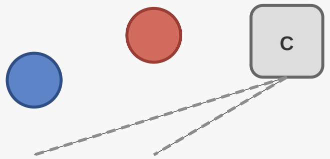
similar actors choose each other

Socialization



connected actors become more alike

Context



a third factor affects both nodes

Speaker notes

Walk through the three causal diagrams: selection (similarity → tie), socialization (tie → similarity), context (third factor causes both). Without temporal data, all three explain the same cross-sectional correlation.

Quick Check 5.B: Designing a Test

You observe that smokers tend to befriend other smokers. You want to determine whether this is selection (smokers prefer smoker friends) or socialization (people start smoking because their friends do).

1. What kind of data would you need beyond a single snapshot of friendships?
2. Sketch one observation that would favor each explanation.

Speaker notes

Give them 4 minutes. (1) Longitudinal data: friendships and smoking status observed at multiple time points. (2) Selection signature: people become friends *after* both already smoke. Socialization signature: people start smoking *after* their friends do. The point is that mechanism identification requires temporal sequencing.

Quick Check 5.B: Solution

Longitudinal observations — friendships and smoking status at multiple time points, ideally with newly formed ties and newly adopted behavior visible.

- : New friendships disproportionately form between people who already share smoking status.
- : People who befriend smokers become more likely to start smoking themselves over the next observation period.

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Quick Check 5.B: Solution

1. Required data: Longitudinal observations — friendships and smoking status at multiple time points, ideally with newly formed ties and newly adopted behavior visible.

- : New friendships disproportionately form between people who already share smoking status.
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Quick Check 5.B: Solution

1. Required data: Longitudinal observations — friendships and smoking status at multiple time points, ideally with newly formed ties and newly adopted behavior visible.

2. Distinguishing observations:

- : New friendships disproportionately form between people who already share smoking status.
- : People who befriend smokers become more likely to start smoking themselves over the next observation period.

Quick Check 5.B: Solution

1. Required data: Longitudinal observations — friendships and smoking status at multiple time points, ideally with newly formed ties and newly adopted behavior visible.

2. Distinguishing observations:

- *Selection*: New friendships disproportionately form between people who already share smoking status.
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1. Required data: Longitudinal observations — friendships and smoking status at multiple time points, ideally with newly formed ties and newly adopted behavior visible.

2. Distinguishing observations:

- *Selection*: New friendships disproportionately form between people who already share smoking status.
- *Socialization*: People who befriend smokers become more likely to start smoking themselves over the next observation period.

Takeaway

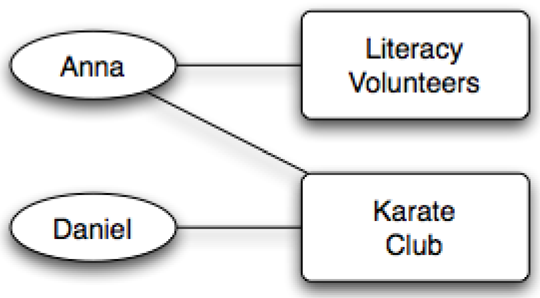
> Cross-sectional graphs rarely settle the causal question. Longitudinal network data is much more informative.

Module 3

Affiliation Networks

> **Guiding Question:** How do we represent the social settings in which ties form?

Bipartite Affiliation



Source: Easley & Kleinberg 2010

Affiliation networks distinguish at least two node types:

- persons
- foci such as groups, locations, classes, or organizations

Formal Definition

An affiliation network is naturally modeled as a bipartite graph:

- one node set for actors
- one node set for foci
- edges only between actors and foci

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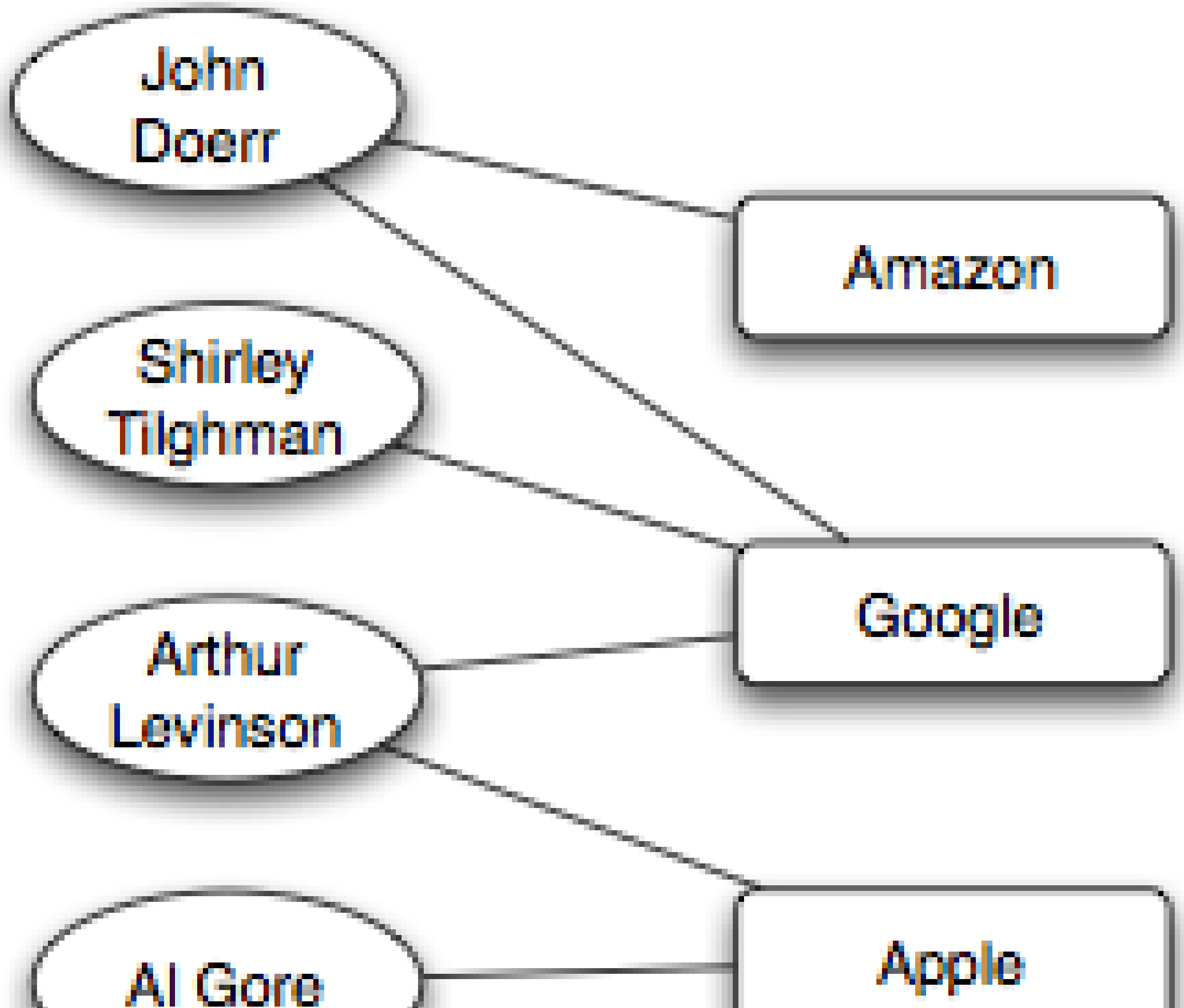
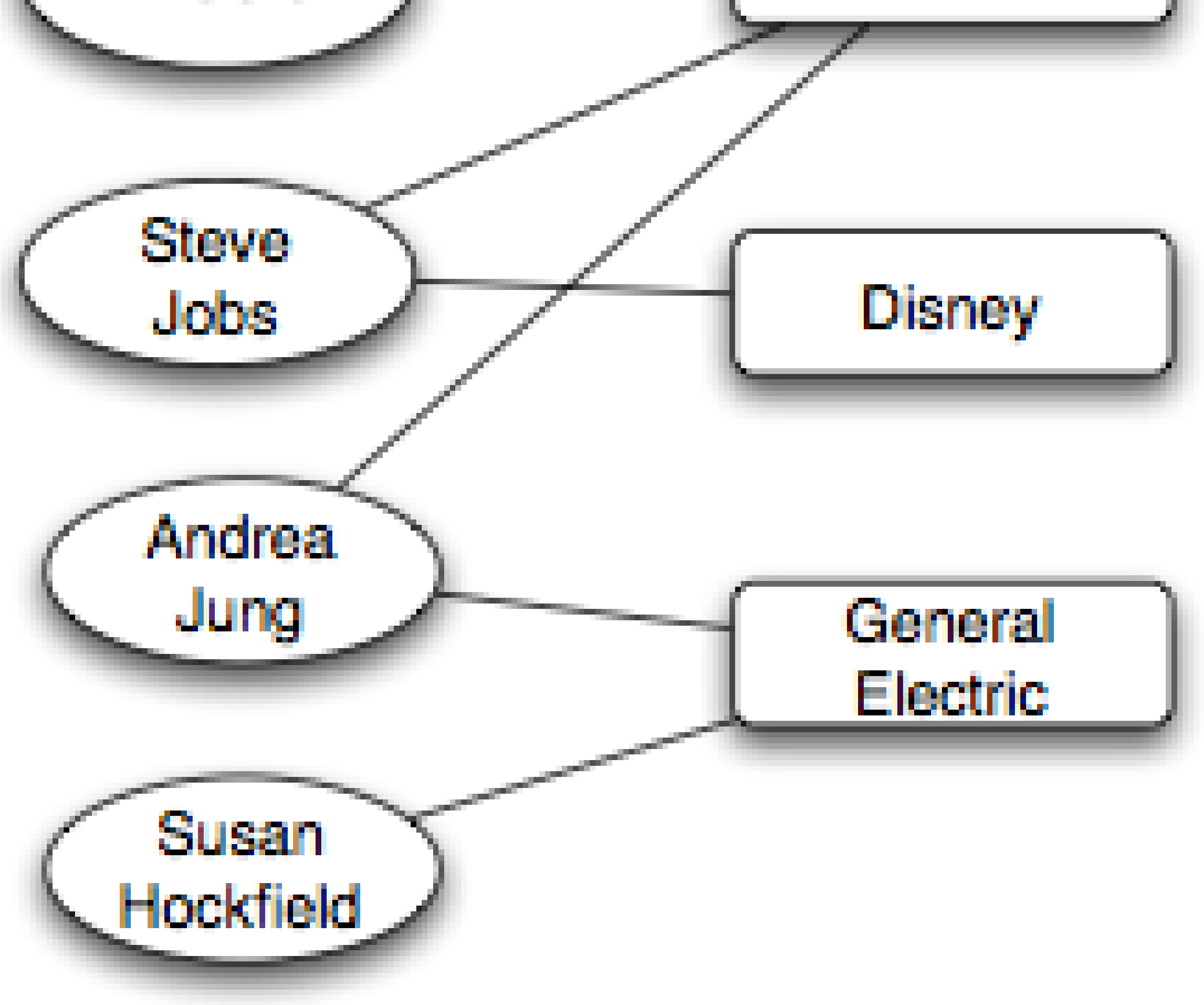
Speaker notes

This slide makes the modeling step explicit. It should connect back to Lecture 02, where bipartite graphs were introduced as an extension of the simple graph model.

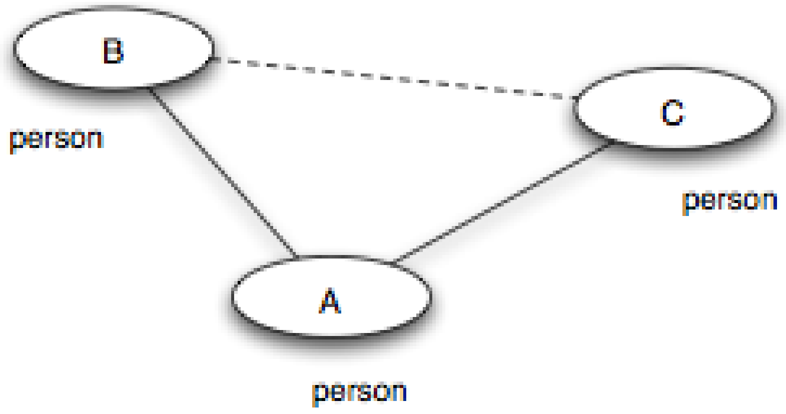
Example


John Doerr
Shirley Tilghman
Arthur Levinson
Al Gore
Amazon
Google
Apple

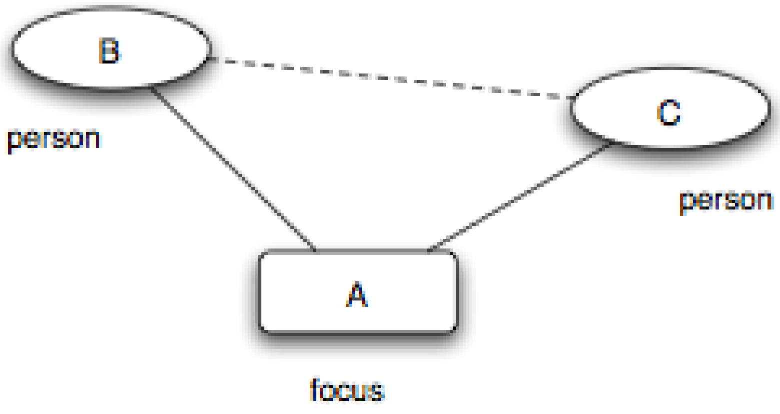


Source: Easley & Kleinberg 2010

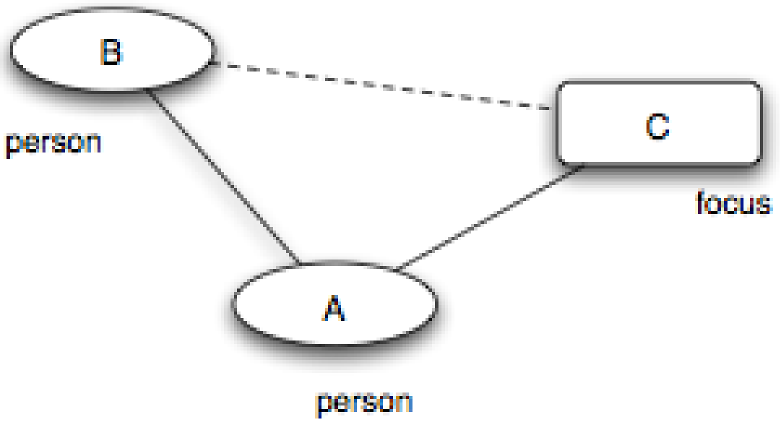
Closure Processes



(a) *Triadic closure*



(b) *Focal closure*



Source: Easley & Kleinberg 2010

Speaker notes

Three closure processes operate in affiliation networks:

- *Triadic closure*: friend-of-friend → friend (person-person).
- *Focal closure*: two persons sharing a focus become friends (person-focus-person).
- *Membership closure*: a friend introduces you to a focus (person-person-focus).

All three are mechanisms that turn structure into more structure.

Quick Check 5.C: Identify the Closure Type

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
3. A friend invites you to their book club, and you join.

Speaker notes

Give them 3 minutes. (1) Focal closure — the elective is the shared focus. (2) Triadic closure — the new friendship forms via a shared friend (the advisor). (3) Membership closure — your friend introduces you to a focus you did not previously belong to.

Quick Check 5.C: Solution

1. — the elective is the shared focus that brings them into contact.
2. — the advisor is the shared friend that closes the triangle.
3. — your friend brings you into a focus you did not previously belong to.

Quick Check 5.C: Solution

1. **Focal closure** — the elective is the shared focus that brings them into contact.
2. — the advisor is the shared friend that closes the triangle.
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Quick Check 5.C: Solution

1. **Focal closure** — the elective is the shared focus that brings them into contact.
2. **Triadic closure** — the advisor is the shared friend that closes the triangle.
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Quick Check 5.C: Solution

1. **Focal closure** — the elective is the shared focus that brings them into contact.
2. **Triadic closure** — the advisor is the shared friend that closes the triangle.
3. **Membership closure** — your friend brings you into a focus you did not previously belong to.

Takeaway

> Affiliation networks make context explicit and let us formalize focal closure and membership closure alongside ordinary triadic closure.

Module 4

Online Link Formation

> **Guiding Question:** What can large online datasets teach us about how links form over time?

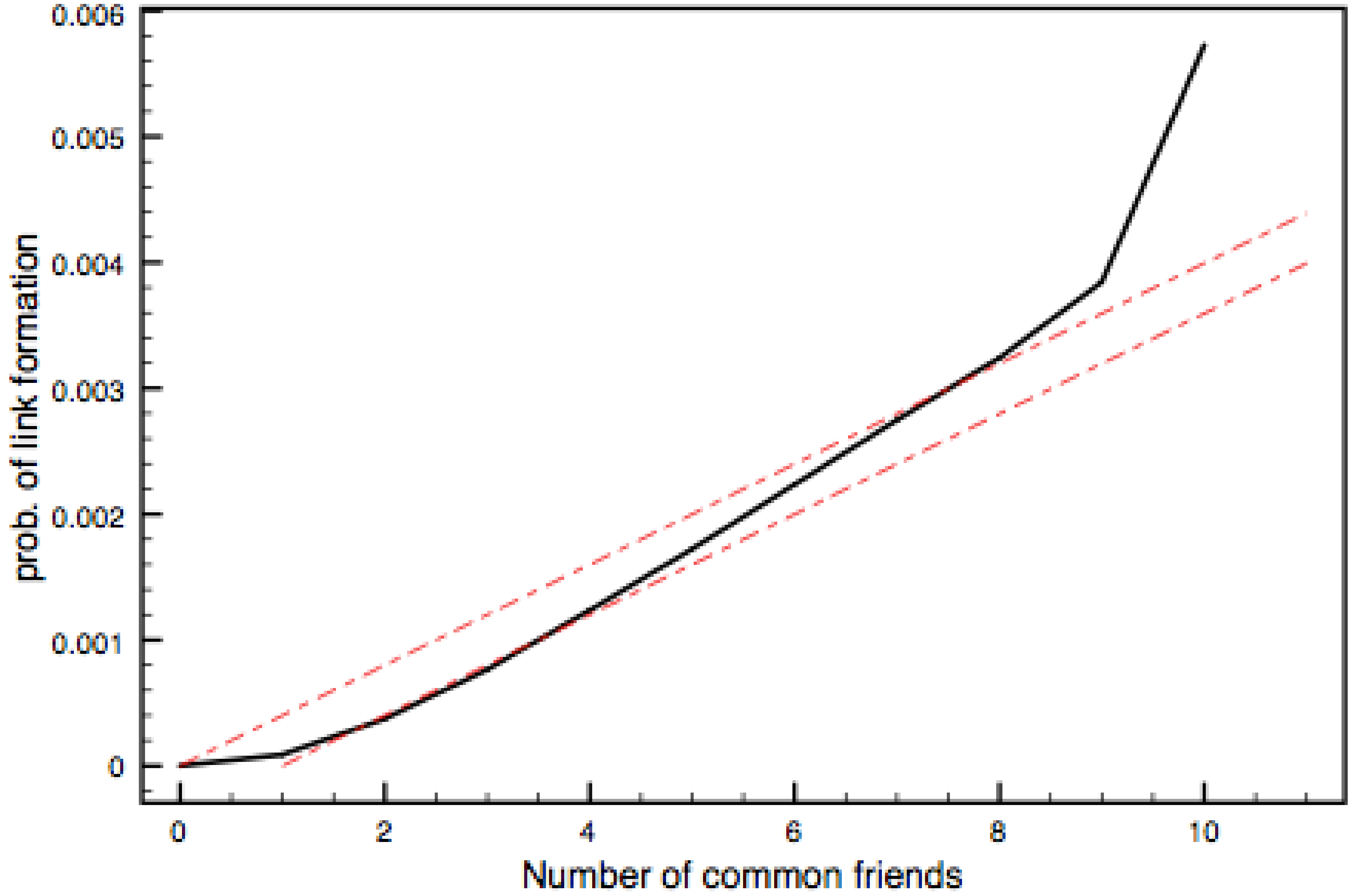
Formal Intuition

Link formation can be studied longitudinally by observing when new edges appear and which local conditions precede them.

Speaker notes

Online platforms make link formation a measurable process. Every new edge has a timestamp, and every preceding state is observable. This is what makes large-scale empirical tests of homophily and closure possible.

Triadic Closure in Email



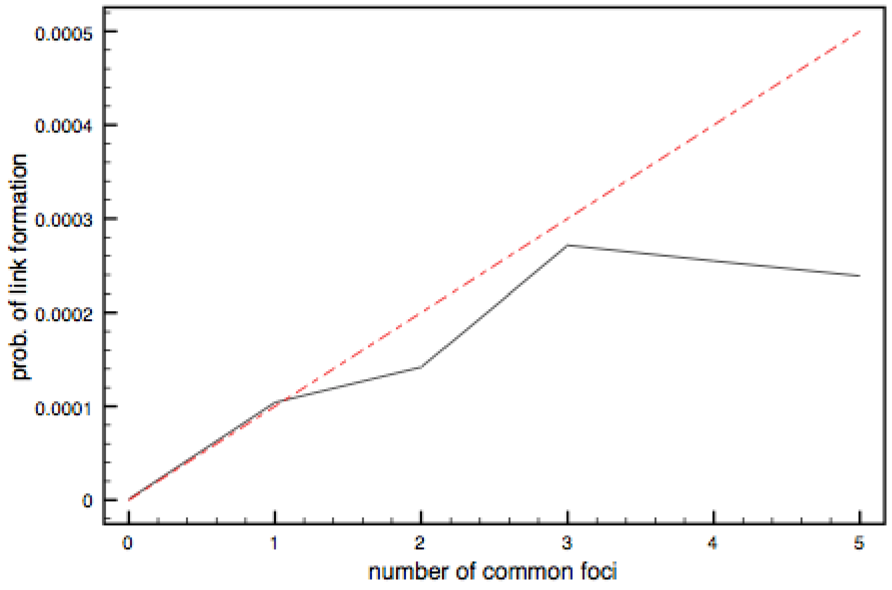
Source: Easley & Kleinberg 2010

The probability of a new edge rises with the number of shared contacts.

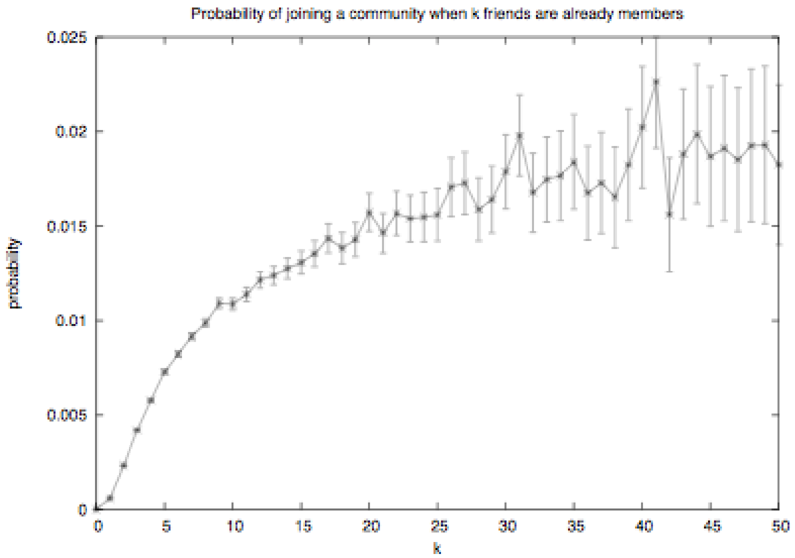
Speaker notes

The empirical curve shows that having 1 shared friend already raises the closure probability significantly above baseline, and additional shared friends amplify the effect. But the curve flattens — diminishing returns rather than linear growth.

Focal and Membership Closure



Source: Easley & Kleinberg 2010



Source: Easley & Kleinberg 2010

Interpretation

Shared foci and shared friends both matter, but often with diminishing returns rather than simple linear growth.

Speaker notes

This is the empirical takeaway: closure is not deterministic. Shared structure raises the probability of a new edge, but always with diminishing marginal returns. No single shared contact "guarantees" closure.

Quick Check 5.D: Reading a Closure Curve

A study finds that the probability of two coworkers becoming Facebook friends rises from 1% (no shared friends) to 20% (one shared friend) to 30% (two shared friends) to 33% (three or more).

1. Describe the shape of the curve.
2. What does the diminishing-returns pattern suggest about the mechanism of closure?

Speaker notes

Give them 3 minutes. (1) Steep initial rise, then flattening. (2) The first shared friend provides most of the information / opportunity / trust. Additional shared friends add less because much of the social-context "credit" has already been collected. The curve is not consistent with a purely additive model.

Quick Check 5.D: Solution

Steep initial rise from 1% to 20% with the first shared friend, then much smaller gains (30%, 33%) for additional shared friends.

Diminishing returns suggest that the first shared contact captures most of the available "social credit" — the new opportunity, the trust, the introduction. Additional shared friends reinforce but do not multiply this effect. The mechanism is not purely additive.

Quick Check 5.D: Solution

1. Shape: Steep initial rise from 1% to 20% with the first shared friend, then much smaller gains (30%, 33%) for additional shared friends.

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Quick Check 5.D: Solution

1. Shape: Steep initial rise from 1% to 20% with the first shared friend, then much smaller gains (30%, 33%) for additional shared friends.

2. Mechanism interpretation: Diminishing returns suggest that the first shared contact captures most of the available "social credit" — the new opportunity, the trust, the introduction. Additional shared friends reinforce but do not multiply this effect. The mechanism is not purely additive.

Takeaway

> Online traces make network formation measurable, but the mechanism still needs interpretation.

Module 5

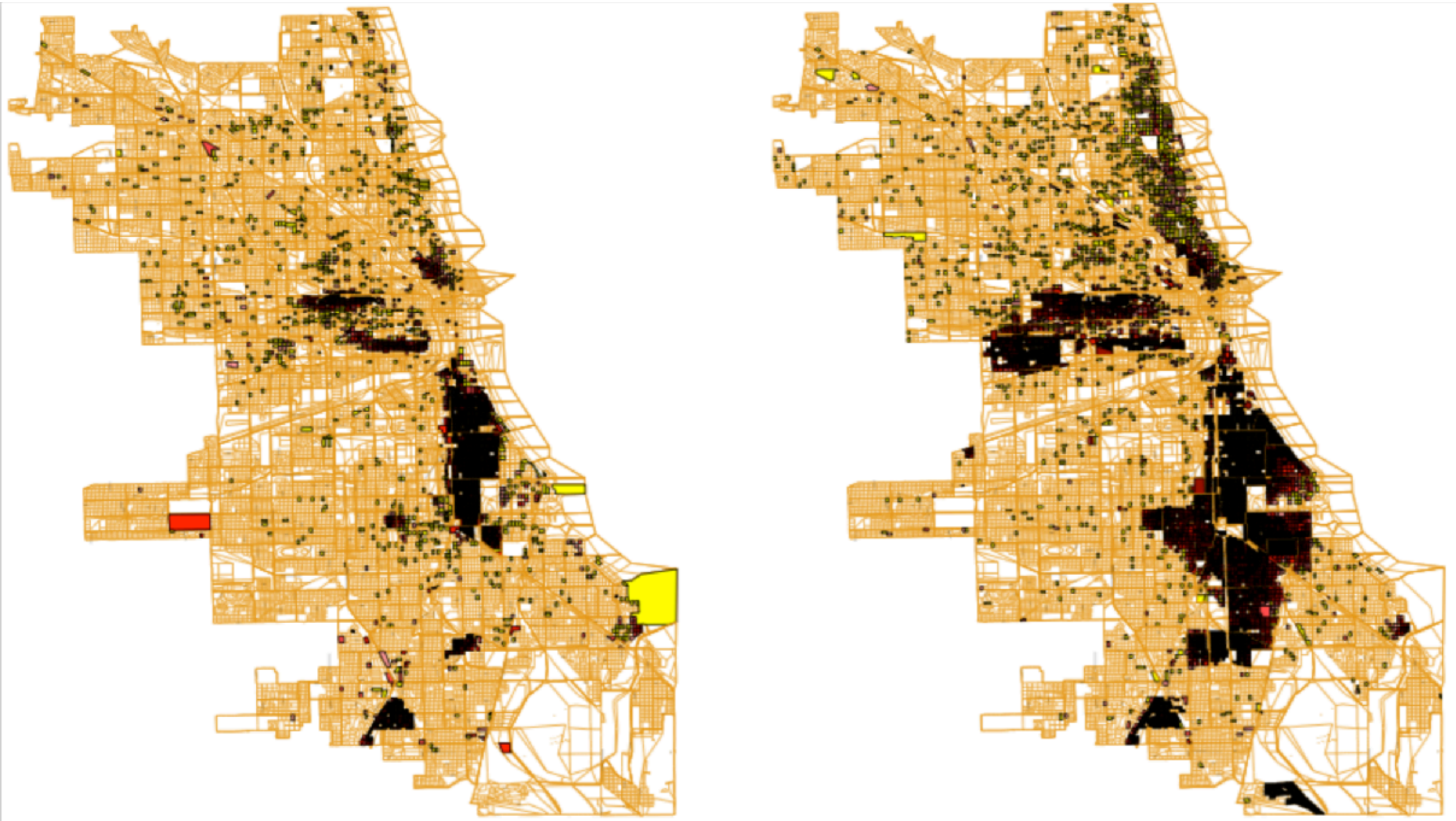
Spatial Segregation

> **Guiding Question:** Can weak local preferences generate strong global segregation?

Formal Intuition

Segregation can emerge from repeated local updates even when each individual preference is mild.

Empirical Motivation



(a) *Chicago, 1940* (b) *Chicago, 1960*

Source: Easley & Kleinberg 2010

Speaker notes

The empirical map raises a puzzle: how could such sharp boundaries emerge if individual preferences are not extreme? Schelling's answer is that even mild preferences, applied repeatedly through local moves, produce sharp global patterns.

Schelling Interpretation

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- agents prefer some minimum level of similar neighbors

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- dissatisfied agents move locally

Schelling Interpretation

- agents prefer some minimum level of similar neighbors
- dissatisfied agents move locally
- many local moves can produce sharply segregated global patterns

Why This Matters

The lesson is not limited to geography. Many network processes transform mild local rules into strong large-scale structure.

Speaker notes

The same logic applies to network homophily, online filter bubbles, and recommendation systems. Mild local preferences + repeated updates = sharp global segregation. No central planner needed.

Quick Check 5.E: Schelling in Networks

Translate Schelling's spatial model into a network setting.

1. What plays the role of "neighbor" in a network rather than a grid?
2. What plays the role of "moving"?
3. Predict: would you expect online recommendation algorithms to amplify or weaken homophily? Why?

Speaker notes

Give them 4 minutes. (1) Network neighbors are graph-adjacent nodes; "neighborhood" is the local subgraph. (2) "Moving" becomes rewiring — dropping ties to dissimilar contacts and adding ties to similar ones. (3) Recommendation algorithms typically amplify homophily because they recommend similar content / people, accelerating the rewiring process toward more similar neighborhoods.

Quick Check 5.E: Solution

A graph-adjacent node — your immediate contacts.

Rewiring — dropping ties to dissimilar contacts and forming new ones with similar contacts.

3. Algorithmic amplification: Recommendation systems typically homophily. They surface similar content and similar people, which accelerates the Schelling-style rewiring toward more similar neighborhoods. The local rule ("recommend what is similar") is mild, but the global outcome (filter bubbles) is sharp.

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Quick Check 5.E: Solution

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Quick Check 5.E: Solution

1. Network neighbor: A graph-adjacent node — your immediate contacts.

2. Moving: Rewiring — dropping ties to dissimilar contacts and forming new ones with similar contacts.

3. Algorithmic amplification: Recommendation systems typically *amplify* homophily. They surface similar content and similar people, which accelerates the Schelling-style rewiring toward more similar neighborhoods. The local rule ("recommend what is similar") is mild, but the global outcome (filter bubbles) is sharp.

Takeaway

> Global polarization can emerge without anyone explicitly aiming for it. Mild local preferences plus repeated local updates produce sharp large-scale structure.

Wrap-Up

- links often reflect both social similarity and social context
- longitudinal data helps separate formation from influence
- simple local preferences can generate strong global patterns

Looking Ahead: Lecture 06

If signs (positive and negative) and not just connections matter, how do whole social structures stabilize or destabilize? Structural balance theory turns the question of friend-and-enemy relations into a graph-theoretic claim with measurable predictions.

Speaker notes

Plant the question for next time. The next conceptual step is to add edge attributes (sign) and ask whether the network as a whole is "consistent" — a property that turns out to have a clean structural characterization.

Image Credits & References

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning About a Highly Connected World. *Cambridge University Press*. https://www.cs.cornell.edu/home/kleinber/networks-book/